Vaani Recovery
A voice agent for loan collections that is polite, multilingual and compliant by design, with a live operations dashboard for the humans who supervise it.

The problem
Collections calls are repetitive, high in volume and tightly regulated. Callers who are rushed or poorly trained risk calling the wrong person, calling at the wrong hour, disclosing a loan to a third party, or applying pressure that breaches fair-practice expectations. Borrowers speak mixed languages such as Hinglish, and people in genuine hardship are often handled the same way as people who simply forgot to pay.

The solution
Vaani phones overdue borrowers, confirms who it is speaking to, states the amount and due date, listens to the reason for the delay, and records a dated promise to pay. Cases involving bereavement, distress, a dispute or a request for a person stop the collection conversation and go to a human officer. Supervisors watch calls, promises and escalations live in a web dashboard.

How it works

Phone Twilio or Exotel	**Gnani Prisma** speech to text, 8 kHz, code-mixed	**Gnani Evon** conversation logic, low temperature, two tools: log promise, escalate	**Gnani Timbre** natural voice with pauses and tone	**Records** Redis call memory, Postgres, live dashboard

Each reply is limited to two or three short sentences and ends with one question, so the call stays fast and natural. Numbers, amounts and dates are spoken as words, and the agent mirrors the customer's language mix.

Compliance built in, not bolted on

- **Identity first.** Nothing about the loan is said until the right person is confirmed; a third party who answers learns nothing.
- **Permitted hours.** Calls run only between 8:00 AM and 7:00 PM IST, checked when a call is placed, in the database and in the agent's instructions.
- **No pressure.** No threats, no invented offers; only the options a manager has approved can be mentioned, and the late fee is stated once, as a fact.
- **Humans for hard cases.** Bereavement, distress, disputes and complaints are handed to an officer, with the reason recorded.
- **Accountability.** Roles (manager, officer, compliance) are enforced by database rules, phone numbers are masked, and every change is written to an audit log.
- **Honest about being virtual.** The agent says it is a virtual assistant and that calls are recorded.

What exists today

Voice backend	FastAPI service that chains the three Gnani models, streams telephony audio, remembers each call, and writes promises and escalations in the background. Tested end to end against simulated Gnani services and a real PostgreSQL database.
Dashboard	Eight screens: overview, live calls, escalation queue, promises to pay, call history, call detail, compliance and settings. Live updates, sign-in with roles, server-side pagination and CSV export. Typecheck, lint (no errors), 42 unit tests and the production build all pass.
Voice demo	Speak into the browser and hear the agent reply, using Prisma for speech to text and Timbre for the voice.

Stack
Python and FastAPI, Redis, PostgreSQL and Supabase (auth, row-level security, realtime), React with TypeScript and Tailwind, Gnani Prisma v3.0, Evon v3.3 and Timbre v3.0, Twilio and Exotel telephony.

Honest limits and next steps

- Not yet run against live Gnani, Twilio or Exotel accounts; message formats were built from public descriptions and need confirming in a pilot.
- No performance or recovery-rate figures are claimed. The first pilot should use consented test numbers and measure promise capture and escalation quality.
- Connect the real loan system, tune the conversation on recorded calls, and have legal and compliance teams review the script and regulations before any real customer is called.

Project link: https://github.com/sasmita-vip/vaani-recovery